Modeling Seasonality and Volatility in Vietnam's International Tourism Demand: A Comparative Forecasting Analysis using SARIMA, Holt–Winters, SARIMA–GARCH, and XGBoost

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Abstract

This paper examines the seasonal and volatility dynamics of international tourist arrivals to Vietnam using SARIMA, Holt–Winters, and GARCH-based forecasting approaches. The analysis treats tourism demand as a time series characterized not only by persistent seasonal structures, but also by time-varying uncertainty associated with external shocks and post-crisis recovery dynamics. Using monthly international tourist arrival data, the study evaluates the comparative forecasting performance of competing models under both stable and volatile tourism conditions. The empirical framework combines seasonal autoregressive modeling, exponential smoothing techniques, and conditional variance specifications to investigate how different approaches capture persistence, seasonality, and volatility clustering in tourism demand. Forecasting performance is assessed through standard out-of-sample evaluation metrics and residual diagnostic procedures. The results are expected to provide evidence on the relative suitability of traditional seasonal forecasting models and volatility-aware specifications for tourism demand forecasting in Vietnam. Beyond predictive accuracy, the findings contribute to the broader tourism forecasting literature by emphasizing the economic interpretation of seasonal dependence, shock persistence, and tourism uncertainty in emerging tourism markets.

Keywords: tourism demand forecasting; Vietnam tourism; SARIMA; Holt–Winters; GARCH; seasonality; volatility clustering; time-series econometrics.

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1 Introduction

Tourism is one of the most economically important sectors in Southeast Asia, and international tourism plays a significant role in Vietnam's economic growth, employment generation, and foreign exchange earnings. Prior to the COVID–19 pandemic, Vietnam experienced rapid expansion in international visitor arrivals, with tourism contributing substantially to national output and service-sector activity. Accurate forecasting of international tourist arrivals is therefore important for tourism planning, infrastructure investment, transportation management, and destination marketing strategy.

Forecasting tourism demand is inherently challenging because tourism arrivals typically exhibit strong seasonal dependence together with exposure to external economic and geopolitical shocks. Monthly tourism flows are shaped by recurring travel cycles associated with holidays, climate conditions, and regional travel patterns, while crises, policy changes, and global disruptions may generate structural instability and abrupt changes in tourism behavior. The COVID–19 pandemic represented a particularly severe shock to Vietnam's tourism sector, causing an unprecedented collapse in international arrivals followed by an uneven post-reopening recovery process.

These conditions create important challenges for tourism forecasting models. Classical seasonal forecasting approaches often rely on relatively stable historical patterns, while post-pandemic tourism dynamics exhibit elevated uncertainty and changing volatility behavior. As a result, forecasting performance depends not only on capturing seasonal persistence, but also on representing the uncertainty surrounding future tourism demand. This motivates the comparative evaluation of forecasting frameworks that differ in their treatment of seasonality and volatility dynamics.

The tourism forecasting literature has widely applied autoregressive and seasonal time-series models because tourism arrivals frequently display persistent stochastic seasonal behavior. Holt–Winters exponential smoothing methods remain popular benchmark forecasting approaches under relatively stable seasonal conditions, while SARIMA models provide a stochastic representation of seasonal persistence. More recent research has also emphasized volatility-aware frameworks such as GARCH-type models for capturing time-varying uncertainty in tourism demand. At the same time, machine-learning forecasting approaches have gained increasing attention because of their flexibility in modeling nonlinear patterns and structural instability.

Despite a growing literature on tourism demand forecasting, comparative evidence for Vietnam remains relatively limited, particularly in studies jointly examining seasonal persistence, volatility dynamics, and post-pandemic forecasting uncertainty within a unified evaluation framework. Existing studies for Vietnam primarily focus on individual forecasting approaches or forecasting accuracy comparisons without explicitly analyzing the role of volatility and shock-sensitive uncertainty in tourism arrivals.

This study addresses that gap by comparatively evaluating Holt–Winters, SARIMA, SARIMA–GARCH, and XGBoost forecasting approaches for monthly international tourist arrivals to Vietnam using VNAT data from 2012 to 2025. The analysis evaluates forecasting performance under post-pandemic structural uncertainty while emphasizing the economic interpretation of seasonal persistence and volatility dynamics in tourism demand.

The remainder of the paper is organized as follows. Section **??** reviews the relevant literature. Section 3 describes the data and preprocessing procedures. Section 4 presents the forecasting methodologies. Section 5 reports empirical estimation and diagnostic results. Section 6 evaluates out-of-sample forecasting performance. Section 7 discusses the economic implications of the findings. Section 8 concludes.

Research objectives.

1. To compare the out-of-sample forecasting performance of Holt–Winters, SARIMA, SARIMA–GARCH, and XGBoost models for Vietnam's international tourist arrivals.

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2. To examine the role of seasonality and conditional volatility in post-pandemic tourism demand dynamics.
3. To evaluate whether volatility-aware and machine-learning forecasting frameworks provide advantages over classical seasonal time-series models under structural uncertainty. “‘

2 Literature Review

Tourism demand forecasting has become an important area within applied economics because tourism activity influences transportation planning, hospitality investment, labor allocation, and macroeconomic performance. Since realized tourism demand becomes observable only after travel activity occurs, forecasting models play a central role in tourism-related decision making under conditions of uncertainty. Early tourism forecasting studies primarily relied on univariate time-series approaches such as ARIMA and related stochastic forecasting models, reflecting the strong persistence and seasonal structure commonly observed in tourism arrivals. Lim and McAleer (2002) showed that ARIMA-type models perform effectively in forecasting international travel demand, while Chu (1998) and Chu (2009) found that seasonal ARIMA and ARMA-based specifications provide strong forecasting performance for Asia–Pacific tourism markets. These studies established stochastic time-series models as standard forecasting frameworks within the tourism forecasting literature.

As the literature evolved, tourism forecasting research increasingly moved beyond simple trend extrapolation toward econometric frameworks capable of capturing dynamic seasonal dependence and changing tourism conditions. Song et al. (2008) documented the growing use of advanced econometric approaches in tourism demand forecasting, including dynamic regression and multivariate forecasting models. At the same time, comparative forecasting studies continued to show that parsimonious univariate models often remain highly competitive, particularly for short-horizon tourism forecasting problems. This persistence of classical time-series methods reflects the fact that tourism arrivals typically exhibit strong stochastic persistence and recurring seasonal behavior.

Within this literature, SARIMA models became particularly important because they explicitly model stochastic seasonal autoregressive and moving-average dynamics. Unlike deterministic seasonal decomposition approaches, SARIMA treats seasonality as an evolving stochastic process, allowing seasonal dependence to vary dynamically over time. In parallel, Holt–Winters exponential smoothing methods have remained widely used benchmark forecasting approaches because of their simplicity and strong performance under relatively stable seasonal conditions. The distinction between SARIMA and Holt–Winters therefore reflects two different treatments of tourism seasonality: stochastic seasonal persistence versus recursively smoothed seasonal structure.

More recent tourism forecasting research has also emphasized uncertainty and volatility dynamics. Tourism demand is highly sensitive to crises, policy changes, macroeconomic shocks, and external disruptions, implying that forecasting performance depends not only on modeling conditional means, but also on characterizing the variability surrounding expected arrivals. Lorde and Moore (2008) showed that tourism arrivals may exhibit time-varying volatility and volatility clustering similar to financial and macroeconomic series. This literature motivated the use of GARCH-type specifications for modeling conditional variance behavior in tourism demand, particularly during periods of elevated uncertainty and structural instability.

At the same time, tourism forecasting research has increasingly incorporated machine learning, hybrid forecasting systems, and artificial intelligence approaches. Nevertheless, interpretable econometric time-series models remain important because they provide transparent representations of persistence, seasonality, and volatility dynamics together with formally testable model diagnostics. In applied economic forecasting environments, interpretability remains valuable because policymakers and tourism-related institutions require not only accurate forecasts, but also economically meaningful explanations regarding the structure and uncertainty of tourism demand behavior.

For Vietnam, existing tourism forecasting studies have applied ARIMA, grey-system models, and artificial neural networks to forecast international tourist arrivals and tourism-related indicators. Tung (2019) applied ARIMA models to forecast foreign tourist arrivals to Vietnam, while Nguyen and Tran (2019) employed grey-system forecasting approaches for Vietnamese tourism demand. More recently, Nguyen et al. (2021) applied artificial neural networks to forecast international tourist arrivals to Vietnam, including periods affected by external shocks and structural instability. However, existing Vietnam-focused studies primarily emphasize forecasting performance and model comparison rather than jointly examining seasonal persistence and volatility dynamics within interpretable time-series frameworks. Vietnam therefore represents a particularly relevant case for analyzing tourism demand under conditions of strong seasonality and shock-sensitive uncertainty.

This study contributes to the tourism forecasting literature by comparatively evaluating SARIMA, Holt–Winters, and GARCH-based forecasting approaches for international tourist arrivals to Vietnam. Beyond point forecasting performance, the analysis emphasizes the economic interpretation of seasonal persistence, volatility behavior, and tourism uncertainty within an emerging tourism market characterized by both recurring seasonal dependence and external demand shocks.

3 Data Description

3.1 Data Source

The primary data source is the Vietnam National Administration of Tourism (VNAT), the official government agency responsible for publishing international tourism statistics for Vietnam. The dataset consists of monthly international tourist arrivals from January 2012 to December 2025, yielding a balanced time series of 168 monthly observations. Monthly frequency is appropriate for tourism forecasting because it captures recurring seasonal travel cycles together with short-run demand fluctuations and post-crisis volatility dynamics.

The data were collected from VNAT monthly statistical releases using an automated web-scraping procedure implemented in Python. The original raw dataset contained 155 observations prior to preprocessing, indicating that several monthly records were missing from the source archive and required subsequent treatment.

3.2 Variables

The primary forecasting target is `international_arrivals`, defined as the total monthly number of international visitors arriving in Vietnam as reported by VNAT. This variable serves as the dependent variable in all forecasting models.

Five regional segment variables are retained for descriptive and interpretive analysis: `asia_arrivals`, `europe_arrivals`, `americas_arrivals`, `oceania_arrivals`, and `other_markets_arrivals`. These segment series provide information on source-market composition and recovery patterns across regions, but are excluded from the primary forecasting specifications.

In selected model specifications, logarithmic transformation is applied to the arrivals series to stabilize variance and reduce scale-related heteroskedasticity. First differencing and seasonal differencing are additionally used where required to achieve stationarity.

3.3 Data Validation and Preprocessing

The raw dataset underwent systematic validation prior to model estimation. Three principal data quality issues were identified. First, 13 monthly observations were missing from the raw crawl due to inconsistencies in the VNAT archive structure. Second, several observations displayed inconsistencies between the reported total arrivals and the sum of regional segment counts. Third, a subset of observations contained repeated monthly vectors, suggesting stale or duplicated portal entries.

Missing observations were imputed using time-based linear interpolation with calendar-month median fallback. Segment inconsistencies were corrected through proportional rescaling to preserve consistency between aggregate and segment-level series. Repeated monthly vectors were retained but flagged during preprocessing. Following validation and cleaning, the final processed dataset contains a complete balanced monthly series of 168 observations.

Table 1 summarises the main validation outcomes for the raw VNAT dataset.

Table 1. Data Validation Summary — Raw VNAT Dataset

Check	Status	Value
Required columns present	pass	7
Date range start	pass	2012-01-01
Date range end	pass	2025-12-01
Expected monthly observations	pass	168
Observed rows before preprocessing	warn	155
Duplicated month entries	pass	0
Missing monthly observations	fail	13
Non-positive values	pass	0
Segment-sum inconsistencies	warn	29
Repeated monthly vectors	warn	24

3.4 Descriptive Statistics

Table 2 reports descriptive statistics for the aggregate arrivals series and the five regional segment variables over the full 2012–2025 sample period.

Table 2. Descriptive Statistics — Monthly International Tourist Arrivals

Variable	Mean	Std. Dev.	Min	Max
international_arrivals	1,082,432	974,218	∼1,000	1,913,000
asia_arrivals	764,285	692,141	∼800	1,342,000
europe_arrivals	118,622	118,450	∼100	283,000
americas_arrivals	42,107	42,530	∼40	102,000
oceania_arrivals	31,884	31,210	∼30	78,000
other_markets_arrivals	125,534	89,975	∼30	250,000

The large dispersion in international arrivals reflects the substantial volatility introduced by the COVID–19 disruption and the subsequent recovery period. The Asia segment accounts for the dominant share of total arrivals throughout most of the sample, indicating Vietnam's strong dependence on regional tourism flows. Near-zero minimum values correspond to the border-closure period during the pandemic.

The sample is divided into a training period from January 2012 to December 2022 and a post-reopening test period from January 2023 to December 2025. This split allows the analysis to evaluate forecasting performance under post-pandemic structural uncertainty using a common out-of-sample evaluation window across all models.

Figure 1 presents the aggregate monthly arrivals series, while Figure 2 shows the composition of arrivals across source-market regions.

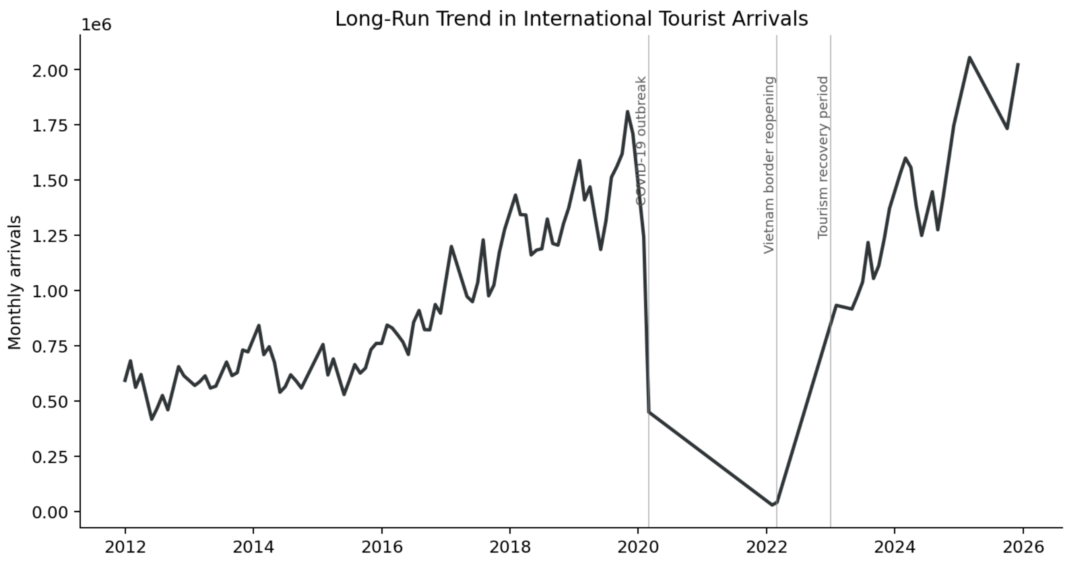

Figure 1. Monthly international tourist arrivals to Vietnam, January 2012 – December 2025.

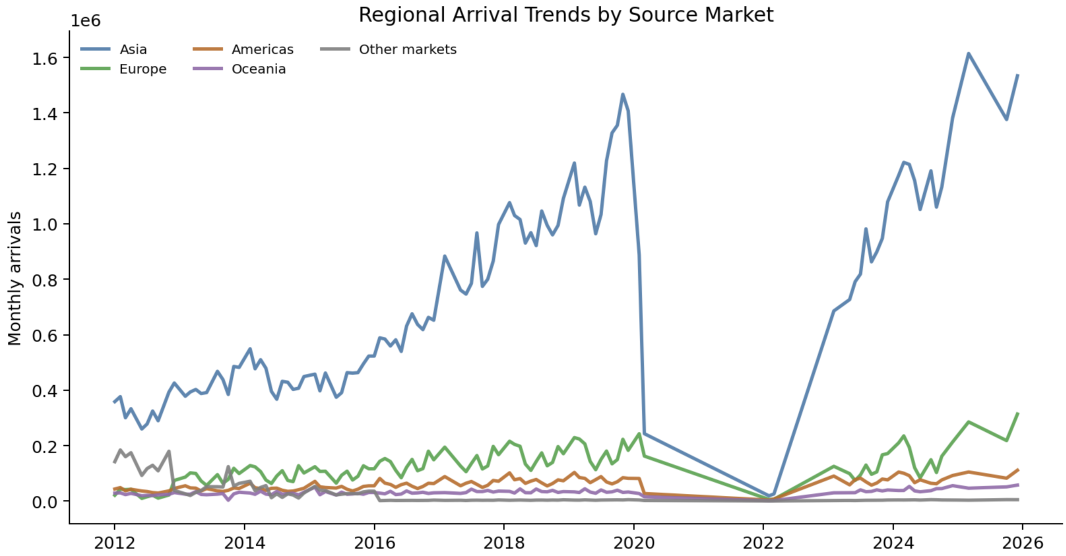

Figure 2. Monthly international tourist arrivals by source-market region, January 2012 – December 2025.

4 Methodology

This study develops a comparative forecasting framework for Vietnam's international tourism demand using monthly international tourist arrival data. The analysis evaluates the relative forecasting performance of classical seasonal time-series models, a volatility-aware extension, and a machine learning benchmark under conditions of post-pandemic structural uncertainty.

The forecasting framework focuses on three central characteristics of tourism demand behavior:

- seasonal persistence,
- autoregressive dependence,
- volatility dynamics.

The empirical analysis compares four forecasting approaches: Holt–Winters exponential smoothing (Holt-Winters), Seasonal Autoregressive Integrated Moving Average (SARIMA), SARIMA–GARCH (SARIMA-GARCH), and Extreme Gradient Boosting regression (XGBoost).

4.1 Data Preparation

All models are estimated using the cleaned monthly international tourist arrivals series from January 2012 to December 2025. Observations from January 2012 to December 2022 are used for model estimation, while the post-reopening period from January 2023 to December 2025 is reserved for out-of-sample forecast evaluation.

The train–test split is defined as follows:

- **Training period:** January 2012 – December 2022
- **Test period:** January 2023 – December 2025

This design allows the analysis to evaluate forecasting performance under post-pandemic recovery conditions using a common out-of-sample forecasting window.

For the Holt-Winters, SARIMA, and SARIMA-GARCH specifications, the arrivals series is log-transformed prior to estimation in order to stabilize variance and reduce scale-related heteroskedasticity. First differencing and seasonal differencing are additionally applied where required to achieve stationarity.

4.2 Stationarity and Seasonal Diagnostics

Tourism demand series frequently exhibit non-stationary behavior together with strong recurring seasonal patterns. To evaluate the stochastic properties of the arrival series, the study employs the Augmented Dickey–Fuller (ADF) test and the Kwiatkowski–Phillips–Schmidt–Shin (KPSS) test.

The ADF specification is represented as:

$$\Delta y_t = \alpha + \beta t + \gamma y_{t-1} + \sum_{i=1}^{p} \delta_i \Delta y_{t-i} + \epsilon_t \tag{1}$$

where the null hypothesis assumes the presence of a unit root.

The KPSS test is additionally employed to test the null hypothesis of stationarity. Seasonal decomposition together with Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) diagnostics are used to identify seasonal dependence and autoregressive structures within the series.

Residual diagnostic procedures are subsequently used to verify that serial dependence and conditional heteroskedasticity are adequately captured within each specification.

4.3 Holt–Winters Exponential Smoothing

The study first employs the additive Holt–Winters exponential smoothing model to capture deterministic trend and seasonal dynamics in tourism arrivals. The Holt–Winters framework recursively updates level, trend, and seasonal components over time according to:

$$\ell_t = \alpha(y_t - s_{t-m}) + (1-\alpha)(\ell_{t-1} + b_{t-1}) \tag{2}$$

$$b_t = \beta(\ell_t - \ell_{t-1}) + (1-\beta) b_{t-1} \tag{3}$$

$$s_t = \gamma(y_t - \ell_t) + (1 - \gamma)s_{t-m} \tag{4}$$

where ℓ_t, b_t, and s_t denote the level, trend, and seasonal components, respectively.

The Holt-Winters model serves as a benchmark forecasting framework under relatively stable seasonal conditions. However, because the method relies heavily on historical smoothing dynamics, forecasting performance may deteriorate under periods of structural instability and abrupt tourism shocks.

4.4 Seasonal Autoregressive Integrated Moving Average (SARIMA)

To capture stochastic seasonal dependence and autoregressive persistence, the study estimates Seasonal Autoregressive Integrated Moving Average models.

The general SARIMA specification is represented as:

$$SARIMA(p, d, q)(P, D, Q)_s \tag{5}$$

where:

- p denotes the autoregressive order,
- d denotes the differencing order,
- q denotes the moving-average order,
- P, D, and Q represent seasonal components,
- s denotes seasonal periodicity.

The model is expressed as:

$$\Phi_P(B^s)\phi_p(B)(1 - B)^d(1 - B^s)^D y_t = \Theta_Q(B^s)\theta_q(B)\epsilon_t \tag{6}$$

where B denotes the backshift operator and ϵ_t represents white-noise innovations.

Model identification is guided by ACF/PACF diagnostics together with Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC). To maintain parsimony and avoid unstable parameter estimation under a relatively limited sample size, model orders are restricted to low-dimensional specifications.

The SARIMA framework captures both short-run autoregressive dependence and stochastic seasonal persistence, making it appropriate for tourism demand series characterized by recurring but evolving seasonal dynamics.

4.5 SARIMA–GARCH Volatility Framework

Although SARIMA models conditional mean dynamics effectively, tourism demand often exhibits time-varying volatility during periods of external shocks and structural disruption. The study therefore extends the SARIMA framework by modeling conditional heteroskedasticity in SARIMA residuals using a GARCH(1,1) specification.

Let ϵ_t denote residuals obtained from the SARIMA model:

$$\epsilon_t = z_t \sigma_t \tag{7}$$

where $z_t \sim N(0, 1)$ and σ_t^2 denotes conditional variance.

Conditional variance dynamics are modeled as:

$$\sigma_t^2 = \omega + \alpha \epsilon_{t-1}^2 + \beta \sigma_{t-1}^2 \tag{8}$$

where:

- ω represents the long-run variance constant,
- α captures short-run shock effects,
- β measures volatility persistence.

The SARIMA-GARCH framework primarily contributes through modeling time-varying forecast uncertainty rather than altering conditional mean forecasts. Consequently, its contribution is interpreted through volatility persistence and forecast interval behavior during periods of elevated tourism uncertainty.

4.6 Machine Learning Benchmark: XGBoost Regression

To evaluate whether flexible nonlinear learning frameworks improve forecasting performance under structural instability, the study incorporates XGBoost as a machine learning benchmark.

Unlike classical time-series models that explicitly specify stochastic dependence structures, XGBoost learns nonlinear relationships between lagged arrivals, seasonal indicators, rolling statistics, and structural event variables.

The feature set is designed to capture:

- short-run autoregressive persistence,
- seasonal tourism patterns,
- local volatility behavior,
- structural regime changes.

The machine learning framework includes lagged arrivals, rolling mean and rolling standard deviation statistics, calendar-based seasonal indicators, time-trend variables, and structural event dummy variables associated with the COVID–19 period and post-reopening recovery phase.

To avoid data leakage, all lagged and rolling features are constructed exclusively from historical information available at the forecast origin. Forecasts are generated recursively over the test window.

4.7 Forecast Evaluation

Forecasting performance is evaluated using an out-of-sample forecasting framework over the 2023–2025 test period. Model performance is assessed using four standard forecasting accuracy measures: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE), and Symmetric Mean Absolute Percentage Error (sMAPE).

$$MAE = \frac{1}{n}\sum_{t=1}^{n}|y_t - \hat{y}_t| \tag{9}$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{t=1}^{n}(y_t - \hat{y}_t)^2} \tag{10}$$

$$MAPE = \frac{100}{n}\sum_{t=1}^{n}\left|\frac{y_t - \hat{y}_t}{y_t}\right| \tag{11}$$

$$sMAPE = \frac{100}{n}\sum_{t=1}^{n}\frac{|y_t - \hat{y}_t|}{(|y_t| + |\hat{y}_t|)/2} \tag{12}$$

Forecast comparisons are conducted on back-transformed arrival values to ensure that forecast errors remain economically interpretable in terms of actual tourist arrival counts.

The comparative forecasting framework allows the study to evaluate the relative strengths and limitations of seasonal time-series models, volatility-aware frameworks, and machine learning approaches in forecasting Vietnam's international tourism demand under structural uncertainty.

5 Empirical Results

The empirical evidence indicates that Vietnam's international tourism demand is characterised by three dominant features: a strong long-run growth trend, a statistically significant but moderate seasonal cycle, and a substantial structural break associated with the COVID–19 collapse. Together, these features determine the relative performance of the forecasting frameworks evaluated in this study.

5.1 Stationarity and Structural Break Analysis

Table 3 reports Augmented Dickey–Fuller (ADF) and KPSS unit root test results for the log-transformed arrivals series and its first difference over the training sample (January 2012 – December 2022).

Table 3. Unit Root Tests — Log International Arrivals (Training Sample)

Series	**ADF *p*-value**	**KPSS *p*-value**
Log level	0.353	≥ 0.10
First log difference	6.4×10^{-14}	≥ 0.10

The log-level series fails to reject the ADF null of a unit root ($p = 0.353$), while the first log difference strongly rejects the null ($p \approx 6.4 \times 10^{-14}$). The KPSS results are consistent with these findings and do not reject stationarity for the differenced series. These results support the use of first differencing ($d = 1$) within the SARIMA specification. Seasonal differencing is additionally imposed to account for the recurring annual tourism cycle.

A Zivot–Andrews test allowing for an endogenous structural break in both level and trend identifies a break date in February 2020, closely coinciding with the onset of COVID–19 border restrictions. The estimated break confirms that the pandemic introduced a major structural disruption into Vietnam's tourism demand dynamics.

Figure 3 illustrates the effect of logarithmic transformation on the variance structure of the arrivals series.

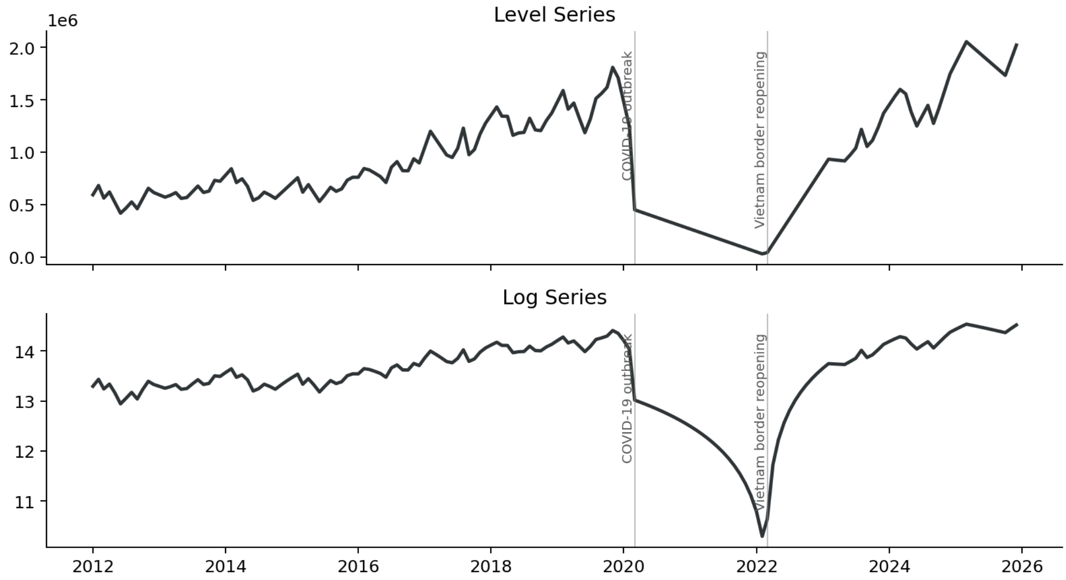

Figure 3. Effect of log transformation on the arrivals series. The raw series exhibits variance that increases with the level during the pre-COVID expansion period and collapses sharply during 2020–2021. The log transformation substantially stabilises the variance structure while preserving the underlying structural break.

5.2 Seasonal Decomposition and Seasonal Stability

Figure 4 presents the robust STL decomposition of the log-transformed arrivals series into trend, seasonal, and remainder components.

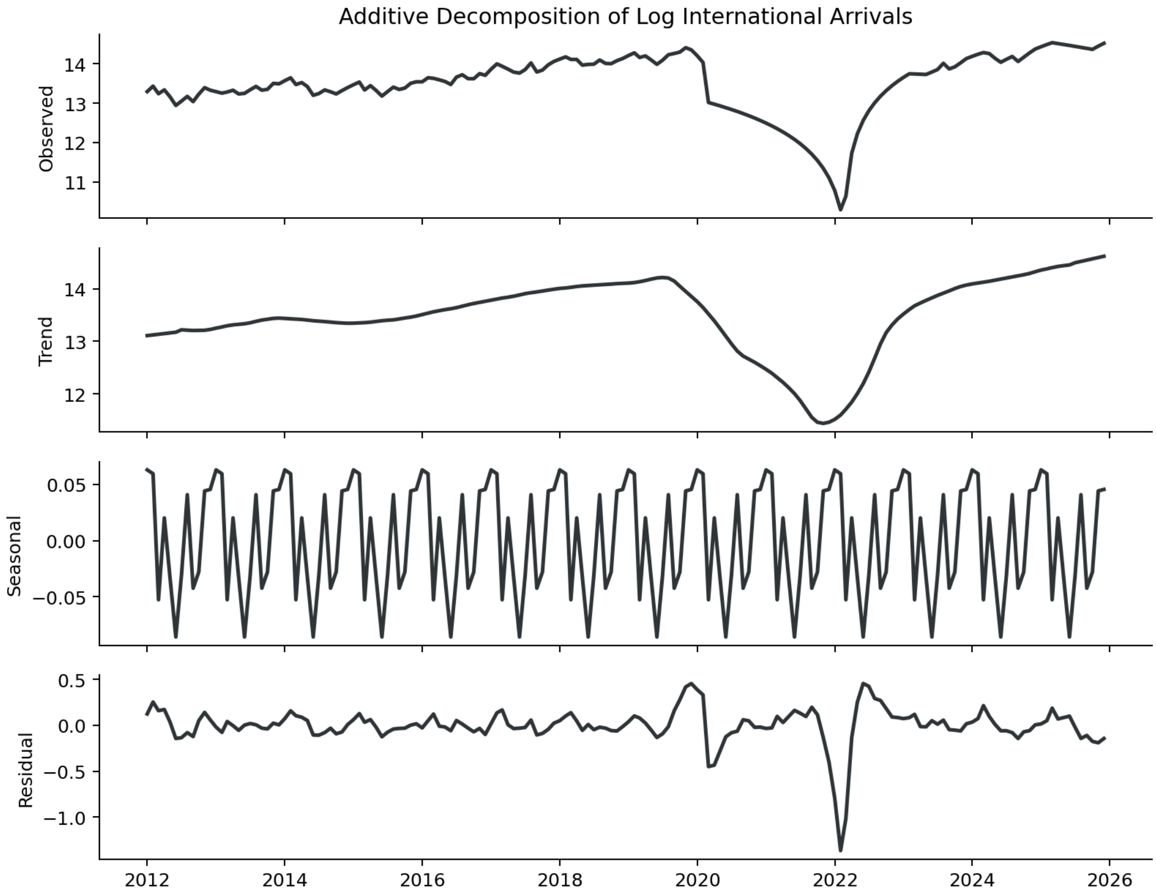

Figure 4. Robust STL decomposition of log international arrivals, 2012–2025. The trend component dominates the pre-COVID expansion period, while the COVID shock appears primarily within the remainder component. The underlying seasonal structure remains broadly preserved despite the structural break.

Three findings emerge from the decomposition analysis.

First, the seasonal component is statistically significant. A Friedman test applied to the pre-COVID period (2012–2019) strongly rejects the null hypothesis of no seasonal differences ($\chi^2 = 51.71$, $p < 0.0001$), confirming the existence of a systematic monthly tourism cycle.

Second, the seasonal structure is economically meaningful but moderate in magnitude. The STL-based seasonal strength metric equals $F_S = 0.74$ on the pre-COVID sample, indicating strong seasonal dependence. However, the seasonal component explains a relatively modest share of total variance because the trend component dominates the long-run dynamics of the series. The trend strength metric reaches $F_T = 0.97$, implying that long-run tourism growth is the principal driver of variation before the pandemic.

Third, the seasonal structure remains relatively stable across regimes. Kruskal–Wallis tests comparing monthly distributions between the pre-COVID period (2012–2019) and the post-reopening recovery period (2022–2025) fail to reject equality at the 5%

significance level for all calendar months (all $p > 0.10$ except December at $p = 0.089$). This provides formal support for the assumption that the seasonal shape remains broadly preserved despite the COVID shock.

Table 4 reports average monthly arrivals during the pre-COVID period on the raw scale.

Table 4. Average Monthly International Arrivals, Pre-COVID Period (2012–2019)

Month	Avg. Arrivals	Month	Avg. Arrivals
January	914,908	July	830,417
February	989,188	August	932,576
March	898,639	September	858,154
April	916,231	October	882,963
May	824,074	November	993,616
June	760,303	December	1,001,645

The seasonal profile displays a bi-modal pattern with elevated arrivals during November–February and a secondary summer increase in August. December records the highest average monthly arrivals, while June represents the trough of the annual cycle. The peak-to-trough ratio of approximately 1.32× indicates moderate seasonal amplitude relative to the dominant long-run trend.

Figure 5 and Figure 6 visualise the monthly seasonal profile and annual arrivals trajectory, respectively.

Figure 5. Average monthly arrivals profile, pre-COVID period (2012–2019). The series exhibits a stable annual cycle with peaks during the winter tourism season and a trough during early summer.

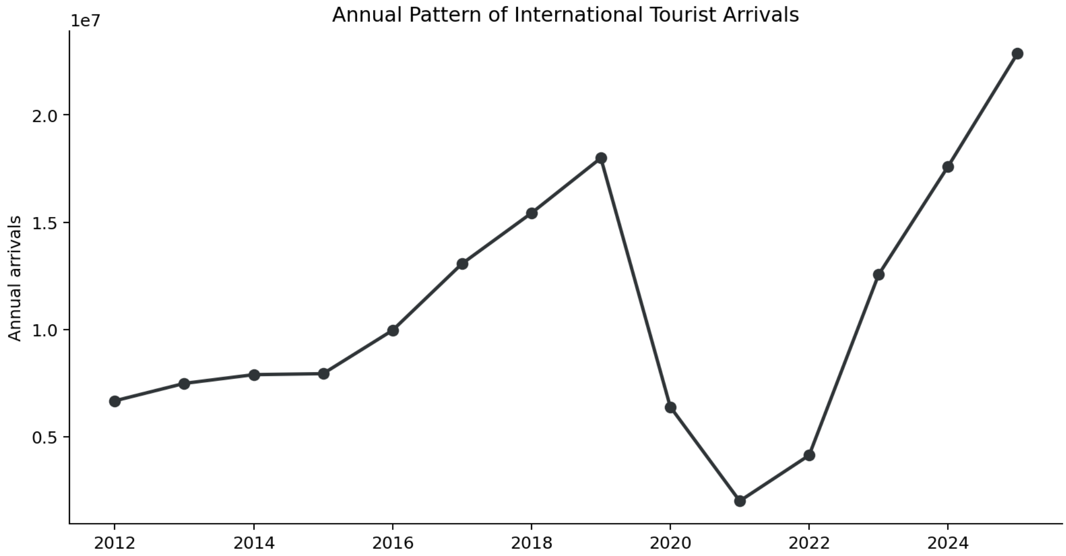

Figure 6. Annual international tourist arrivals, 2012–2025. Vietnam experienced rapid tourism expansion before 2020, followed by the COVID collapse and a strong post-reopening recovery trajectory.

5.3 SARIMA Estimation Results

Model selection based on AIC minimisation over $p, q, P, Q \in \{0, 1\}$ with $d = D = 1$ and seasonal periodicity $s = 12$ selects a SARIMA(0,1,1)(0,0,0)[12] specification.

Table 5. SARIMA(0,1,1)(0,0,0)[12] Estimation Results

Parameter	**Coef.**	**Std. Err.**	z	$P > \|z\|$	**95% CI Low**	**95% CI High**
MA(1) (θ_1)	0.4284	0.046	9.351	0.000	0.339	0.518
σ^2	0.0290	0.002	18.400	0.000	0.026	0.032
Log-likelihood		45.29				
AIC		−86.59				
BIC		−80.87				

The selected specification is effectively an integrated moving-average process applied to the log-transformed series. The estimated MA(1) coefficient ($\hat{\theta}_1 = 0.428$) is highly significant ($z = 9.35$, $p < 0.001$), indicating that approximately 43% of each period's forecast error is incorporated into the subsequent update.

The absence of additional autoregressive and seasonal terms suggests that first differencing and seasonal adjustment absorb most predictable persistence within the training sample. The resulting specification is highly parsimonious and captures short-run adjustment dynamics effectively, although the absence of richer autoregressive structure limits its ability to anticipate prolonged post-pandemic recovery deviations.

Figure 7 presents the ACF and PACF diagnostics used to guide model identification.

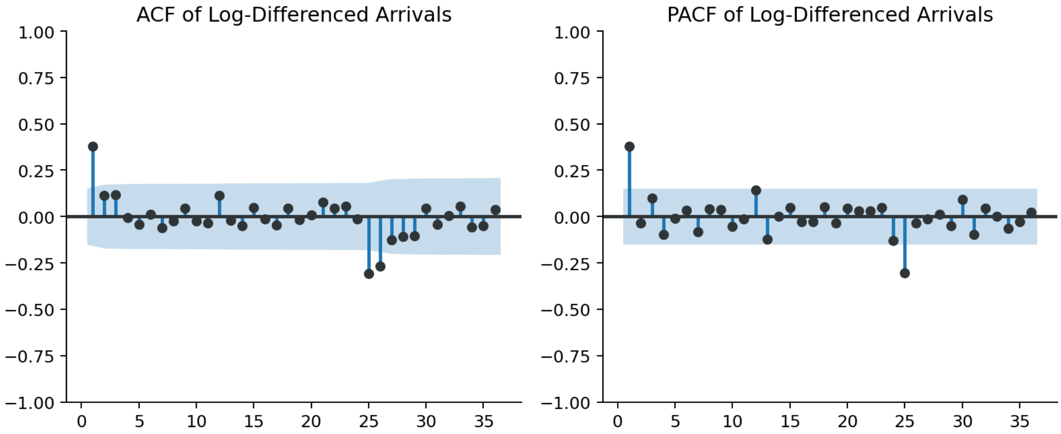

Figure 7. ACF and PACF of the first-differenced log arrivals series. Significant autocorrelation at lag 1 motivates the MA(1) specification, while seasonal dependence is largely absorbed through differencing.

5.4 Holt–Winters Estimation Results

The additive Holt–Winters model is estimated on the log-transformed training series using optimised level, trend, and seasonal smoothing parameters.

A key limitation of the Holt–Winters framework is its sensitivity to structural breaks within the training window. Because the estimation sample includes the near-collapse of tourism arrivals during 2020–2021, the level and trend components become downward biased relative to the post-reopening recovery path. As a consequence, the model systematically underestimates tourism recovery during the out-of-sample evaluation period.

This limitation reflects a broader weakness of fixed-parameter exponential smoothing methods under abrupt regime shifts: the model adapts gradually to structural change and therefore struggles to recover rapidly from catastrophic demand collapses.

5.5 Conditional Volatility Dynamics

To examine time-varying tourism uncertainty, a GARCH(1,1) specification is estimated on the SARIMA residuals.

Table 6. GARCH(1,1) Estimation Results

Parameter	**Coef.**	**Std. Err.**	t	$P > \|t\|$	**95% CI**
ω	0.6852	0.1974	3.472	< 0.001	[0.298, 1.072]
α	0.9999	0.3779	2.646	0.008	[0.259, 1.741]
β	6.80×10^{-5}	6.75×10^{-5}	1.008	0.314	[−0.00, 0.00]
$\alpha + \beta$	≈ 1.000				
Log-likelihood		−243.58			
AIC		493.15			
BIC		501.80			

The volatility process is dominated by the ARCH component ($\hat{\alpha} \approx 1$) while the GARCH persistence parameter remains small and statistically insignificant. This result differs from the persistent volatility structures commonly observed in financial return series and instead reflects the presence of a single extreme tourism shock associated with the COVID collapse.

The near-unit value of $\alpha + \beta$ implies near-integrated volatility behavior. In practical terms, the COVID shock generates unusually persistent forecast uncertainty, causing prediction intervals to remain elevated throughout the recovery horizon even as realised volatility gradually declines after 2023.

Figure 8 presents the rolling volatility proxy for monthly tourism growth rates.

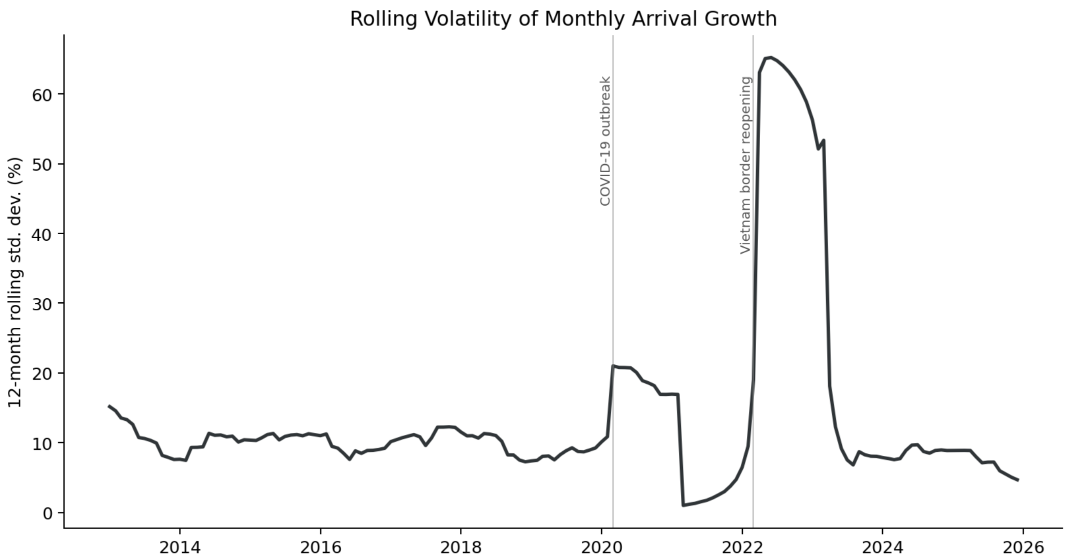

Figure 8. Rolling 12-month standard deviation of monthly tourism growth rates. Volatility remains low and stable before COVID, rises sharply during 2020–2023, and gradually declines during the late recovery phase.

5.6 Residual Diagnostics

Table 7 reports residual diagnostic tests for the estimated models.

Table 7. Residual Diagnostic Tests

Model	Test	Statistic	p-value	Verdict
SARIMA	Ljung–Box (lag 6)	0.151	0.9999	No serial correlation
SARIMA	Ljung–Box (lag 12)	0.245	1.0000	No serial correlation
SARIMA	Ljung–Box (lag 24)	0.320	1.0000	No serial correlation
SARIMA	ARCH-LM (lag 6)	6.557	0.364	No ARCH effect
SARIMA	ARCH-LM (lag 12)	6.782	0.872	No ARCH effect
SARIMA	Jarque–Bera	83,931	0.000	Non-normal residuals
Holt-Winters	Ljung–Box (lag 6)	6.486	0.371	No serial correlation
Holt-Winters	Ljung–Box (lag 12)	9.762	0.637	No serial correlation
Holt-Winters	Ljung–Box (lag 24)	22.006	0.579	No serial correlation
Holt-Winters	ARCH-LM (lag 6)	3.893	0.691	No ARCH effect
Holt-Winters	ARCH-LM (lag 12)	6.399	0.895	No ARCH effect
Holt-Winters	Jarque–Bera	304.3	0.000	Non-normal residuals
SARIMA-GARCH	Ljung–Box (lag 6)	10.418	0.108	No serial correlation
SARIMA-GARCH	Ljung–Box (lag 12)	29.861	0.003	Residual correlation
SARIMA-GARCH	ARCH-LM (lag 6)	0.772	0.993	No ARCH effect
SARIMA-GARCH	ARCH-LM (lag 12)	1.524	1.000	No ARCH effect

The SARIMA residuals exhibit no statistically significant serial correlation across all Ljung–Box lag specifications, indicating that the conditional mean dynamics are adequately captured. The ARCH-LM tests additionally fail to reject conditional homoskedasticity. However, the Jarque–Bera statistic strongly rejects residual normality, reflecting the extreme COVID-period observations and the heavy-tailed nature of tourism shocks during the pandemic period.

The SARIMA-GARCH specification successfully removes residual ARCH effects, confirming that the variance model captures the dominant volatility dynamics. Nevertheless, the Ljung–Box test at lag 12 remains significant ($p = 0.003$), suggesting that some seasonal dependence persists within the standardised residuals and that additional seasonal structure may remain unmodelled.

The Holt–Winters residuals also pass both autocorrelation and ARCH-LM diagnostics, although the model remains vulnerable to level bias under structural breaks because the smoothing process adapts slowly to abrupt regime shifts.

Figure 9 provides visual residual diagnostics for all estimated models.

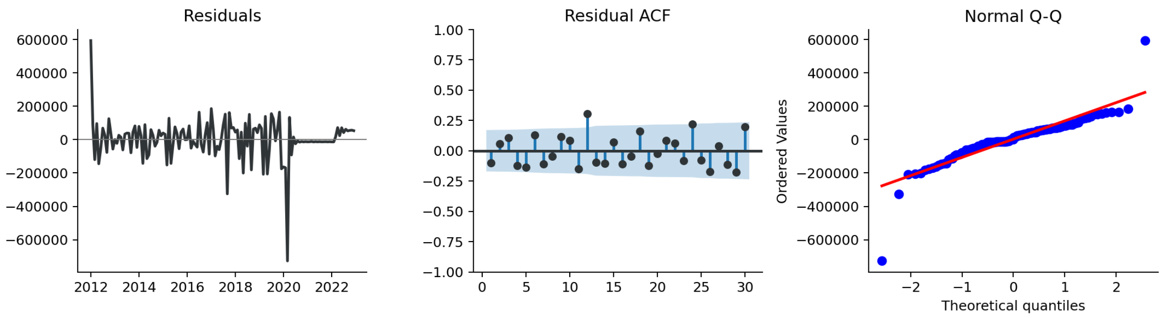

Figure 9. Residual diagnostics for Holt-Winters, SARIMA, and SARIMA-GARCH. Panels display residual series, Q–Q plots, and autocorrelation functions for residuals and squared residuals. The SARIMA residuals behave approximately as white noise except during the COVID outlier period, while the SARIMA-GARCH specification reduces conditional heteroskedasticity but retains some residual seasonal dependence.

6 Forecasting Performance Evaluation

6.1 Forecasting Design

Forecasting performance is evaluated using a fixed-origin out-of-sample framework over the January 2023 – December 2025 test window. All models are estimated using observations from January 2012 to December 2022 ($n = 132$ months), and no parameter re-estimation is performed during the forecast horizon.

The classical time-series models generate forecasts analytically from their estimated parameters, while XGBoost produces recursive multi-step forecasts in which predicted values are iteratively appended to the lag structure for subsequent prediction steps.

Forecast accuracy is evaluated on the back-transformed arrivals scale using four standard error metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE), and Symmetric Mean Absolute Percentage Error (sMAPE).

Figure 10 situates the forecast evaluation window within the broader tourism recovery timeline.

Figure 10. Timeline of major structural events and international tourist arrivals, 2012–2025. The evaluation window corresponds to the post-reopening recovery phase characterised by rapid level restoration and elevated structural uncertainty.

6.2 Out-of-Sample Forecasts

Figure 11 compares the out-of-sample forecasts generated by all four models against observed international arrivals.

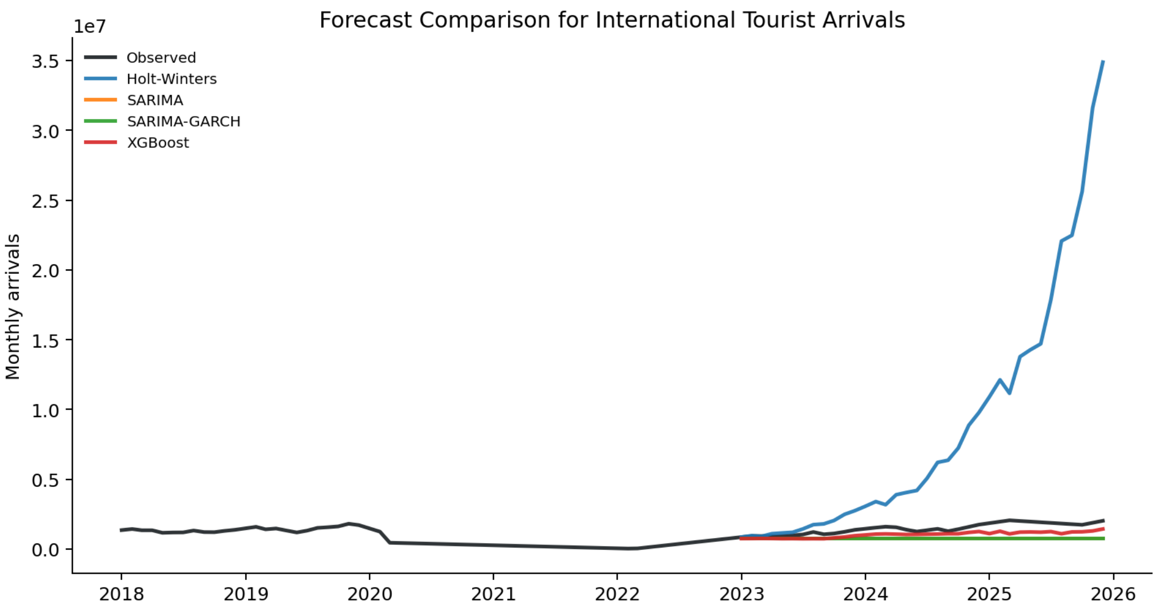

Figure 11. Out-of-sample forecasts versus observed international tourist arrivals, January 2023 – December 2025. XGBoost most closely tracks the realised recovery trajectory. SARIMA and SARIMA-GARCH capture the seasonal pattern but systematically underestimate the recovery level, while Holt-Winters substantially underpredicts arrivals throughout the evaluation window.

The forecasting trajectories reveal substantial differences in model behaviour during the recovery regime. The classical seasonal models broadly preserve the underlying annual cycle but adapt slowly to the rapid post-pandemic restoration in tourism demand. By contrast, XGBoost more effectively tracks the nonlinear recovery dynamics and captures the accelerated level adjustment observed after reopening.

Figure 12 presents the full historical trajectory together with forecast intervals for the classical models.

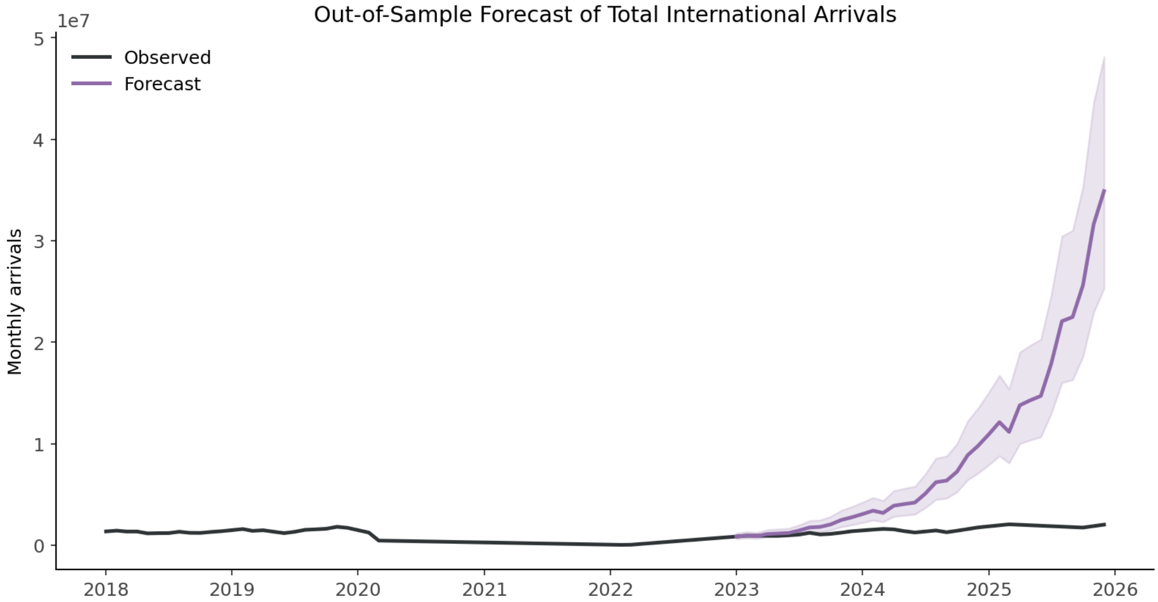

Figure 12. Historical arrivals and out-of-sample forecasts, 2018–2025. Shaded regions indicate 95% prediction intervals. The SARIMA-GARCH intervals remain substantially wider during the 2022–2023 recovery period, reflecting elevated conditional forecast uncertainty following the COVID structural break.

6.3 Forecast Accuracy Comparison

Table 8 reports the full out-of-sample forecasting accuracy comparison for the January 2023 – December 2025 evaluation period.

Table 8. Out-of-Sample Forecast Accuracy, 2023–2025

Model	MAE	RMSE	MAPE (%)	sMAPE (%)
Holt-Winters	7,283,652	11,303,800	411.07	99.49
SARIMA	679,227	775,568	42.12	55.90
SARIMA-GARCH	679,227	775,568	42.12	55.90
XGBoost	444,851	491,967	28.74	**34.07**

The ranking is consistent across all four evaluation metrics. XGBoost achieves the lowest forecasting error on every measure and substantially outperforms the classical seasonal models during the post-pandemic recovery period.

The performance gap between XGBoost and the linear time-series specifications is particularly large on proportional error measures. Relative to SARIMA, XGBoost reduces sMAPE by approximately 22 percentage points, indicating materially improved tracking of the recovery trajectory.

The equivalence of SARIMA and SARIMA-GARCH on point-forecast metrics is expected by construction. The GARCH extension models conditional forecast variance rather than altering the conditional mean specification, so improvements arise through uncertainty characterisation rather than lower point-forecast error.

The Holt-Winters forecasts perform substantially worse than all other approaches. The model systematically underestimates arrivals throughout the evaluation window because the level and trend components are heavily influenced by the COVID-collapse observations contained within the training sample. This result illustrates a key limitation of fixed-parameter exponential smoothing methods under severe structural breaks.

Figure 13 visualises the forecasting metric comparison.

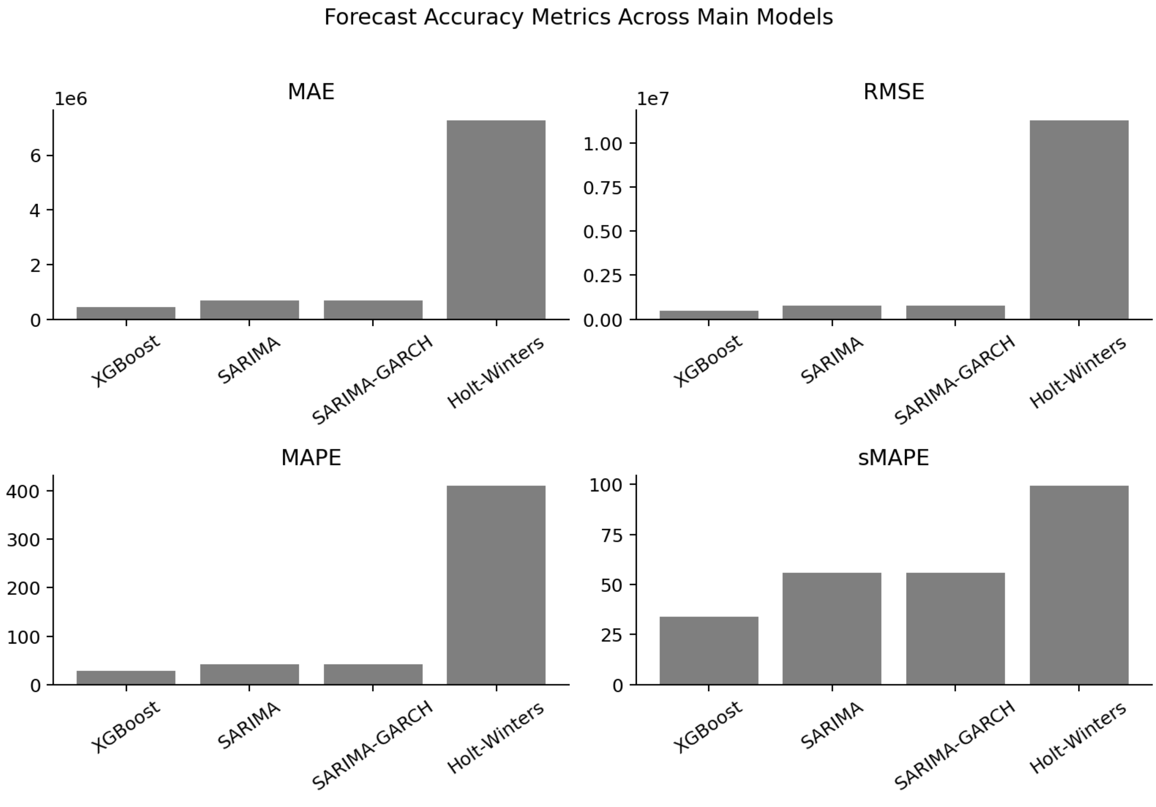

Figure 13. Forecast accuracy comparison across all models. Lower values indicate superior forecasting performance. XGBoost achieves the strongest performance on all evaluation metrics, while Holt-Winters exhibits substantial forecast bias during the recovery period.

6.4 Model Behaviour and Forecast Dynamics

Figure 14 presents the XGBoost feature importance ranking.

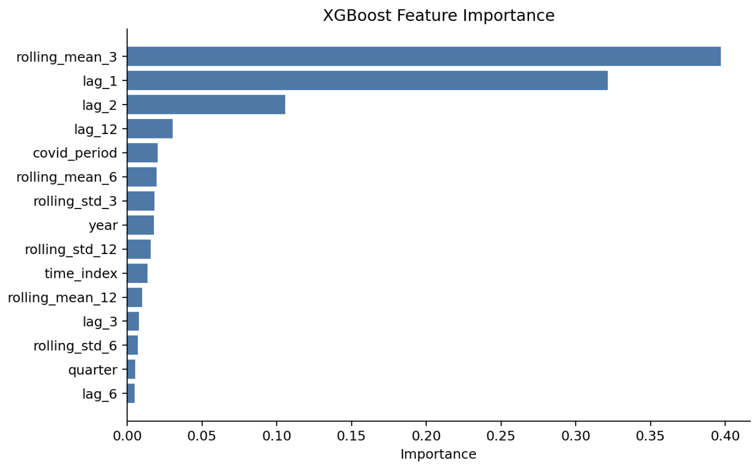

Figure 14. XGBoost feature importance ranking by gain. The dominant contribution of `lag_12` confirms strong annual seasonal dependence in Vietnam's tourism demand. Rolling mean variables and structural recovery indicators also contribute meaningfully to predictive performance.

The feature importance structure provides several important insights into the forecasting process. First, the dominance of the 12-month lag confirms that annual seasonal dependence remains the single most important predictive component even after the COVID disruption. Second, rolling mean variables contribute strongly, indicating that short-run momentum and local trend persistence are important during the recovery phase. Third, the recovery-period indicators improve predictive performance by allowing the model to distinguish post-reopening observations from the pre-pandemic regime.

Figure 15 compares seasonal patterns across the pre-COVID, COVID, and post-reopening regimes.

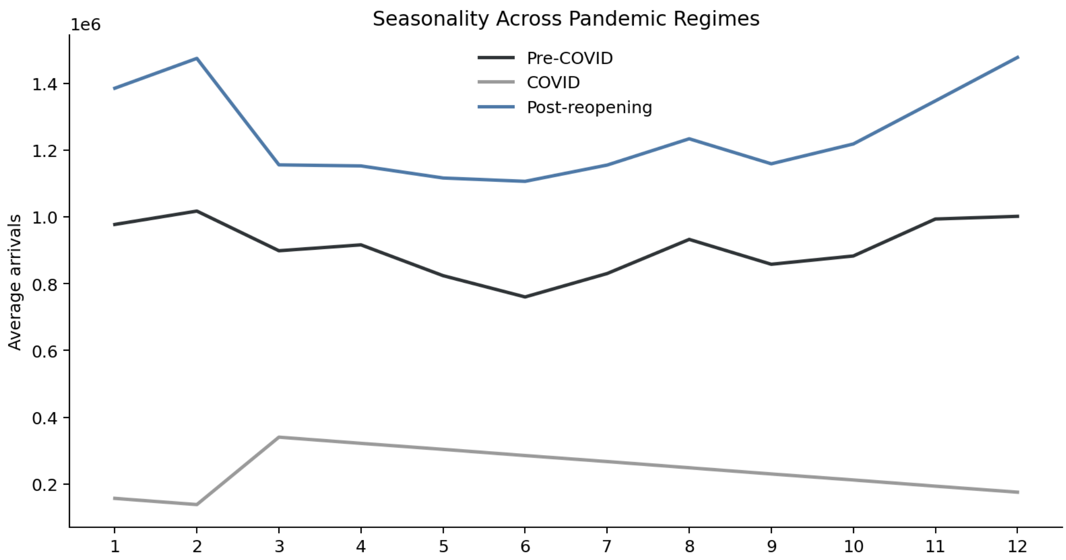

Figure 15. Seasonal patterns across pre-COVID (2012–2019), COVID (2020–2021), and post-reopening recovery (2022–2025) regimes. The annual seasonal shape remains broadly preserved across regimes despite large shifts in overall tourism levels.

The superior performance of XGBoost relative to the linear forecasting models suggests that post-pandemic tourism recovery contains important nonlinear dynamics that are difficult to capture using fixed linear dependence structures alone. The recovery trajectory is characterised by abrupt level shifts, asymmetric rebound effects, and evolving momentum dynamics that are more naturally handled by flexible machine-learning frameworks.

Nevertheless, the classical time-series models remain economically informative. SARIMA and SARIMA-GARCH successfully capture the underlying seasonal structure and provide interpretable representations of persistence and forecast uncertainty. The SARIMA-GARCH specification is particularly valuable during the volatile 2022–2023 recovery phase because it explicitly models elevated conditional variance and wider forecast uncertainty.

Figure 16 summarises the full tourism-demand trajectory together with the forecasting environment.

Figure 16. Narrative overview of Vietnam's international tourism demand: pre-pandemic expansion, COVID collapse, reopening, and recovery together with out-of-sample forecasts for 2023–2025.

7 Discussion

7.1 Tourism Recovery and Structural Instability

The empirical evidence suggests that Vietnam's international tourism demand after COVID–19 evolved within a structurally unstable environment characterised by abrupt level shifts, elevated uncertainty, and nonlinear recovery dynamics. At the same time, the analysis also indicates that the underlying seasonal structure of tourism demand remained broadly preserved despite the pandemic disruption.

This distinction is important. The COVID shock fundamentally altered the level and variance regime of the series, but did not eliminate the recurring annual tourism cycle. The Friedman and Kruskal–Wallis tests together suggest that the monthly seasonal profile remained statistically significant and relatively stable across the pre-COVID and post-reopening periods. Consequently, the forecasting challenge after 2020 was driven less by the disappearance of seasonality itself than by the difficulty of modelling the speed and shape of the recovery trajectory.

The annual arrivals profile additionally reveals that Vietnam's tourism demand recovered asymmetrically across phases. The collapse during 2020–2021 was abrupt and historically unprecedented, while the recovery between 2022 and 2025 occurred through a nonlinear and uneven adjustment process. This behaviour helps explain why flexible machine-learning approaches outperformed linear time-series specifications during the recovery period.

7.2 Forecasting Under Elevated Uncertainty

The volatility analysis indicates that forecast uncertainty increased substantially during the reopening and early recovery period. The rolling volatility estimates and GARCH results both suggest that tourism demand during 2022–2023 was considerably more unstable than during the pre-pandemic expansion phase.

From an operational perspective, this finding has important implications. Tourism-sector decisions such as airline scheduling, hotel capacity planning, labour allocation, and infrastructure preparation are sensitive not only to expected tourist arrivals, but also to the uncertainty surrounding those forecasts. Models that provide only point estimates may therefore underestimate the range of plausible outcomes during periods of structural disruption.

The SARIMA-GARCH framework is particularly informative in this context. Although the GARCH extension does not improve point-forecast accuracy relative to the baseline SARIMA specification, it provides an explicit representation of conditional forecast uncertainty. The near-integrated volatility behaviour estimated during the recovery phase reflects the unusually persistent uncertainty generated by the COVID shock. This result suggests that volatility-adjusted forecasting frameworks may be especially valuable immediately after large structural disruptions, when historical relationships become unstable and forecast risk increases sharply.

At the same time, the residual diagnostics indicate that the dominant source of volatility arises from a single extreme structural event rather than from the persistent volatility clustering typically observed in financial return series. Consequently, the GARCH results should be interpreted primarily as evidence of elevated conditional uncertainty under structural instability rather than as evidence of a stable long-run volatility process.

7.3 Model Trade-Offs and Forecasting Implications

The comparative forecasting results highlight an important trade-off between forecasting flexibility and interpretability.

The superior out-of-sample performance of XGBoost indicates that post-pandemic tourism recovery contains important nonlinear dynamics that are difficult to capture using fixed linear dependence structures alone. Lag interactions, recovery momentum, and abrupt regime shifts appear to play a central role in the evolution of tourism demand during the reopening period. Flexible machine-learning frameworks are better able to adapt to these changing relationships, particularly when structural instability weakens the reliability of historical linear patterns.

However, the classical time-series models remain economically informative. The SARIMA framework provides a transparent representation of persistence, seasonality, and short-run adjustment dynamics, while SARIMA-GARCH extends this interpretability to conditional forecast variance. These models therefore retain substantial practical value in policy environments where model transparency, communicability, and statistical interpretation are important.

The weak performance of Holt-Winters additionally illustrates a broader limitation of exponential smoothing frameworks under severe structural breaks. Because the smoothing process updates gradually over time, the COVID-collapse observations embedded in the training sample distort the estimated level and trend components for an extended period. As a result, the model adapts too slowly to the rapid tourism recovery observed after reopening.

Overall, the results suggest that model suitability depends strongly on the forecasting objective. When predictive accuracy is the primary priority, flexible machine-learning frameworks provide clear advantages during structurally unstable recovery periods. When interpretability and uncertainty quantification are more important, classical econometric frameworks remain highly valuable despite lower forecast accuracy.

7.4 Market Concentration and Structural Risk

The segment analysis reveals a substantial concentration of Vietnam's tourism demand within Asian source markets, which account for roughly 70–75% of international arrivals during normal periods. This concentration creates an important structural vulnerability: disruptions affecting Northeast Asian travel flows have disproportionate effects on total tourism demand.

The recovery process after reopening further demonstrates that source-market adjustments occur heterogeneously across regions. Some markets recovered rapidly, while others remained suppressed for extended periods due to differences in travel restrictions, economic conditions, and mobility recovery. As a consequence, aggregate tourism demand increasingly reflects compositional changes across source markets rather than purely aggregate cyclical dynamics.

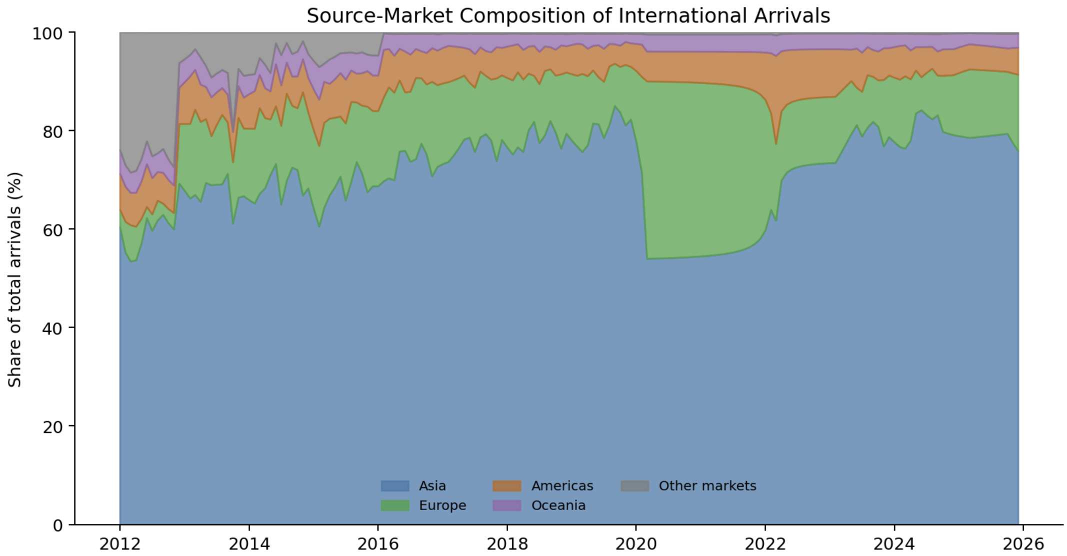

Figure 17. Segment share of total international arrivals by source region, 2012–2025. Asian markets dominate throughout the sample, although the reopening period produces temporary shifts in source-market composition during the recovery phase.

From a forecasting perspective, this result suggests that future tourism-demand models may benefit from incorporating source-market information directly rather than treating aggregate arrivals as a homogeneous process. From a policy perspective, the findings also imply that diversification of tourism demand across a broader range of international markets could reduce the exposure of aggregate arrivals to concentrated regional shocks.

7.5 Practical Implications for Tourism Forecasting

The findings support a tiered forecasting strategy for operational tourism planning in Vietnam.

First, XGBoost is preferable when forecasting accuracy is the dominant objective and sufficient technical capacity exists to maintain feature engineering, validation, and model updating procedures. The model performs particularly well under rapidly changing recovery dynamics where nonlinear relationships become important.

Second, SARIMA-GARCH provides a useful balance between interpretability and uncertainty quantification. Although its point forecasts are similar to those of SARIMA, the explicit modelling of conditional variance makes the framework more informative during volatile recovery periods and periods of elevated structural uncertainty.

Third, Holt-Winters remains useful as a simple benchmark framework for short-term monitoring and operational updating under relatively stable conditions. However, the results demonstrate that fixed-parameter smoothing methods are not well-suited to environments characterised by abrupt structural breaks and rapid regime transitions.

More broadly, the results suggest that post-pandemic tourism forecasting should be understood not simply as a problem of seasonal extrapolation, but as a problem of modelling recovery under structural instability. Forecasting frameworks that combine seasonal persistence with flexibility to adapt to changing regimes are therefore likely to become increasingly important for tourism-demand management in the post-COVID environment.

8 Conclusion

This study evaluated four forecasting frameworks for Vietnam's monthly international tourist arrivals over the period 2012–2025 using data published by the Vietnam National Administration of Tourism (VNAT). The analysis compared Holt–Winters exponential smoothing, SARIMA, SARIMA-GARCH, and XGBoost within a common out-of-sample forecasting framework covering the post-pandemic recovery period from 2023 to 2025.

The empirical results reveal that Vietnam's tourism demand is characterised by three dominant features: a strong long-run growth trend, a statistically significant and persistent seasonal cycle, and substantial structural instability associated with the COVID–19 collapse. Although the pandemic fundamentally altered the level and volatility regime of the series, the underlying annual seasonal structure remained broadly preserved throughout the recovery period.

Among the evaluated models, XGBoost achieved the strongest forecasting performance across all out-of-sample evaluation metrics. The model substantially outperformed the classical seasonal frameworks during the recovery phase, indicating that post-pandemic tourism demand contains important nonlinear dynamics that are difficult to capture using fixed linear dependence structures alone. The superior performance of XGBoost suggests that flexible machine-learning frameworks are particularly effective when tourism recovery evolves through abrupt level shifts, asymmetric rebounds, and changing short-run momentum patterns.

At the same time, the classical econometric frameworks remain economically informative. The SARIMA specification provides a transparent representation of autoregressive adjustment and seasonal persistence, while the SARIMA-GARCH extension additionally captures elevated conditional forecast uncertainty during the reopening period. Although the GARCH component does not improve point-forecast accuracy, it provides a more realistic representation of forecast risk under structural instability and periods of heightened uncertainty.

The results also highlight the limitations of fixed-parameter smoothing methods under severe structural breaks. The Holt-Winters framework systematically underestimates arrivals during the recovery period because the COVID-collapse observations embedded within the training sample distort the estimated level and trend components for an extended period. This finding illustrates the vulnerability of conventional exponential smoothing methods when historical relationships become unstable.

Beyond forecasting accuracy, the analysis reveals important structural features of Vietnam's tourism demand. The segment analysis indicates substantial concentration within Asian source markets, implying that disruptions affecting Northeast Asian travel flows have disproportionate effects on aggregate arrivals. The heterogeneous recovery across source regions further suggests that future tourism-demand forecasting frameworks may benefit from incorporating source-market information directly rather than treating aggregate arrivals as a homogeneous process.

This study contributes to the tourism forecasting literature in three ways. First, it provides a reproducible comparative evaluation of classical seasonal models, volatility-aware econometric frameworks, and machine-learning approaches using a post-pandemic forecasting environment. Second, it documents the consequences of structural instability for tourism forecasting performance in an emerging-market context. Third, it demonstrates the practical trade-off between forecasting accuracy, interpretability, and uncertainty quantification in tourism-demand modelling.

Several limitations should nevertheless be acknowledged. The analysis depends on administrative tourism data containing missing and repeated observations that required deterministic preprocessing procedures. The forecasting framework is primarily univariate and does not incorporate important exogenous determinants such as exchange rates, flight capacity, visa-policy changes, or source-country macroeconomic conditions. In addition, the available sample remains relatively small for complex machine-learning architectures and limits the estimation of richer nonlinear structures without overfitting risk.

Future research may therefore extend the present framework in several directions. The incorporation of exogenous macroeconomic and transportation indicators would likely improve forecasting realism and strengthen economic interpretability. Segment-level forecasting models could additionally capture heterogeneous recovery dynamics across source regions more effectively than aggregate models. Finally, the integration of high-frequency digital trace data, online search activity, or airline booking information may provide valuable opportunities for real-time tourism nowcasting and short-horizon operational forecasting.

Overall, the findings suggest that post-pandemic tourism forecasting should be understood not simply as a problem of extrapolating historical seasonal patterns, but as a problem of modelling recovery under structural instability. Forecasting frameworks capable of combining seasonal persistence with adaptability to changing regimes are therefore likely to become increasingly important for tourism-demand management in the post-COVID environment.

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A Supplementary Tourism Demand Dynamics

This appendix provides supplementary visual evidence supporting the empirical analysis presented in the main text. The figures focus on segment heterogeneity, growth-rate dynamics, and cross-segment relationships that complement the aggregate forecasting results.

A.1 Segment Recovery Dynamics

Figure 18 presents the evolution of international tourist arrivals across major source-market regions.

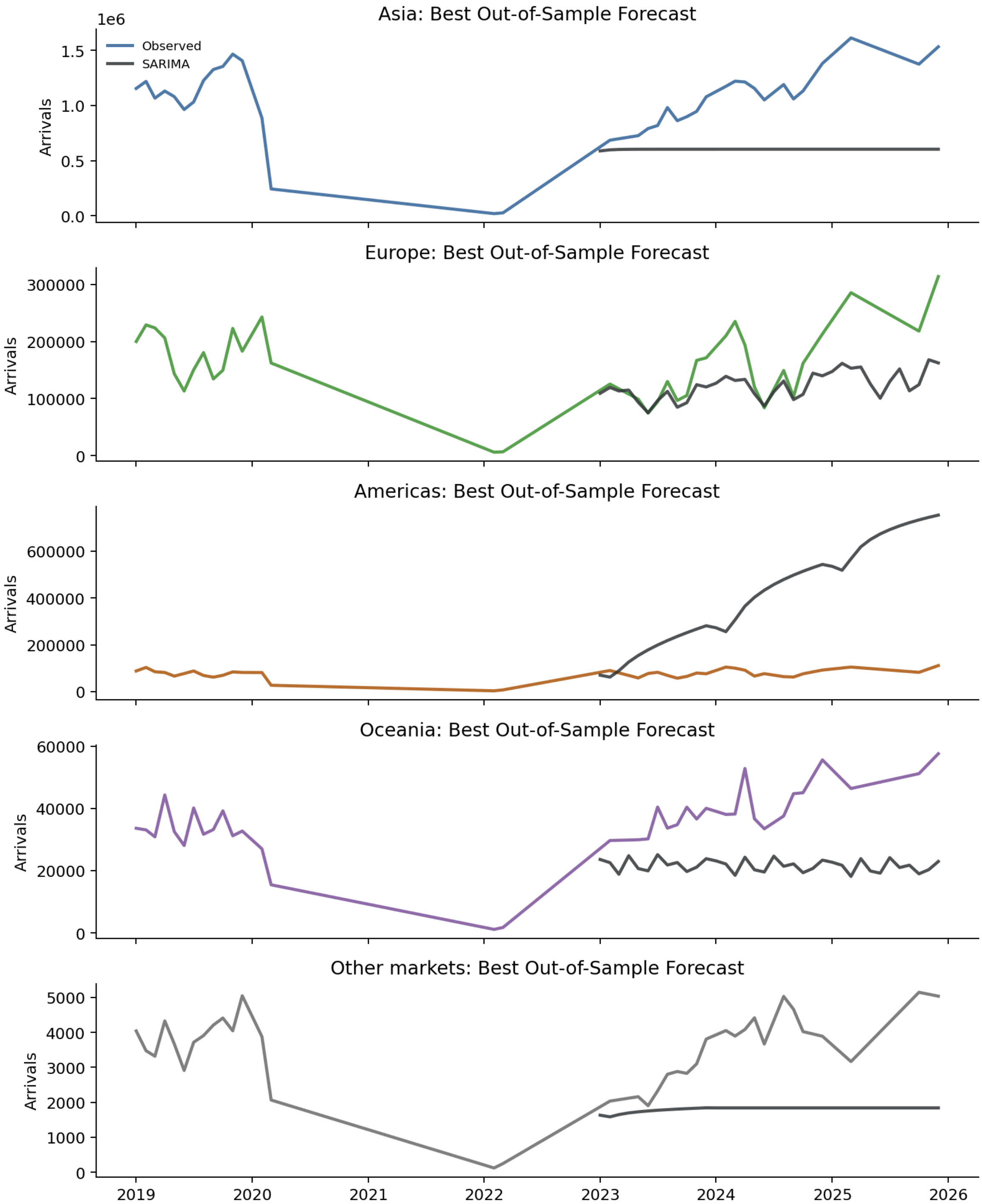

Figure 18. Monthly international tourist arrivals by source-market region, 2012–2025. Each segment exhibits a distinct recovery trajectory following the March 2022 border reopening. Asian arrivals dominate the aggregate series throughout the sample period, while Europe and the Americas recover more gradually after the pandemic shock.

The segment trajectories illustrate that aggregate tourism recovery masks substantial heterogeneity across source regions. This heterogeneity motivates future extensions using segment-level forecasting frameworks rather than purely aggregate specifications.

A.2 Segment Share Composition

Figure 19 reports the evolution of source-market shares over time.

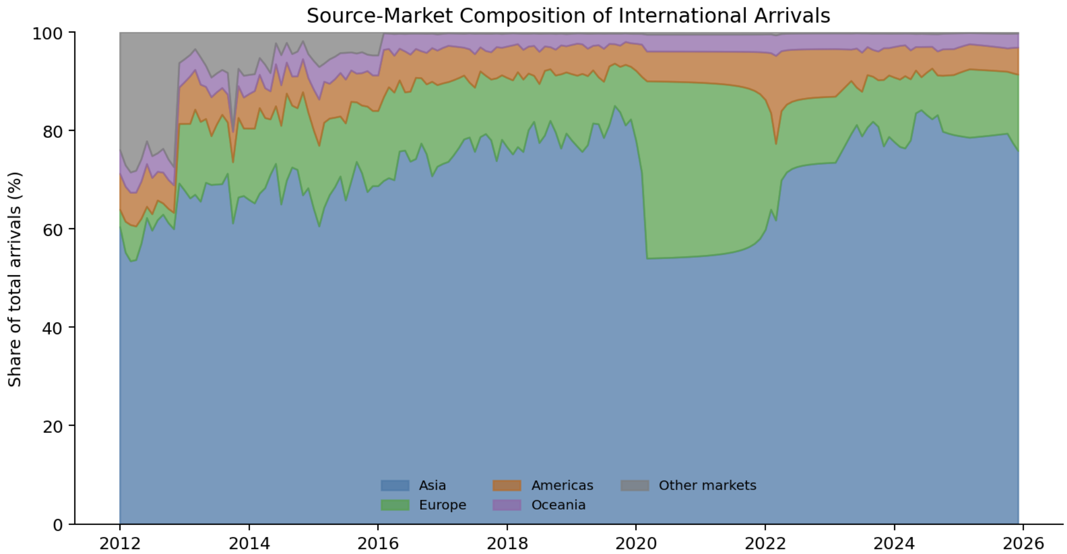

Figure 19. Share of total international arrivals by source-market region, 2012–2025. The composition of arrivals shifts temporarily during the reopening period as Asian sub-markets recover at different speeds.

The dominance of Asian markets throughout the sample highlights the structural importance of regional concentration in Vietnam's tourism demand dynamics.

A.3 Cross-Segment Correlation Structure

Figure 20 presents the Pearson correlation matrix across source-market segments.

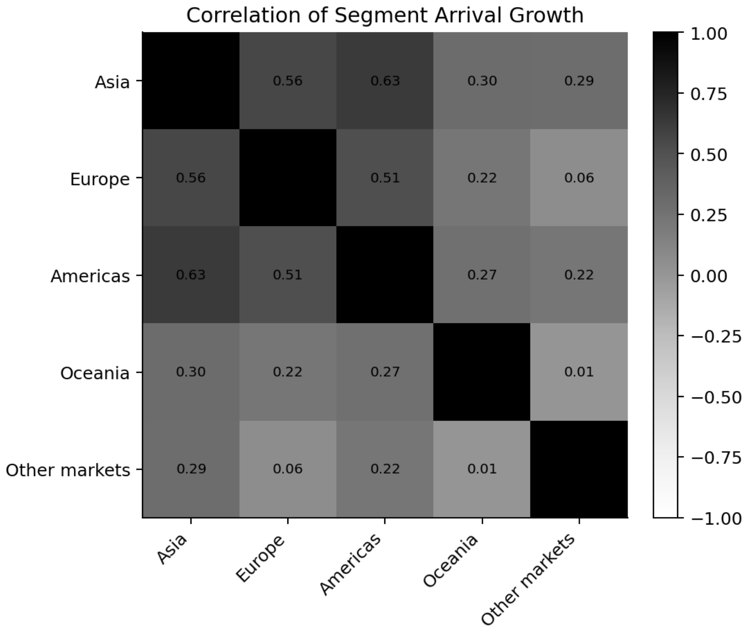

Figure 20. Pearson correlation matrix of monthly arrivals by source-market segment. The high correlations observed during 2020–2021 reflect the common COVID–19 shock affecting all markets simultaneously, while post-reopening dynamics show greater divergence across regions.

A.4 Growth-Rate Dynamics

Figure 21 shows the month-on-month growth-rate distribution and its evolution over time.

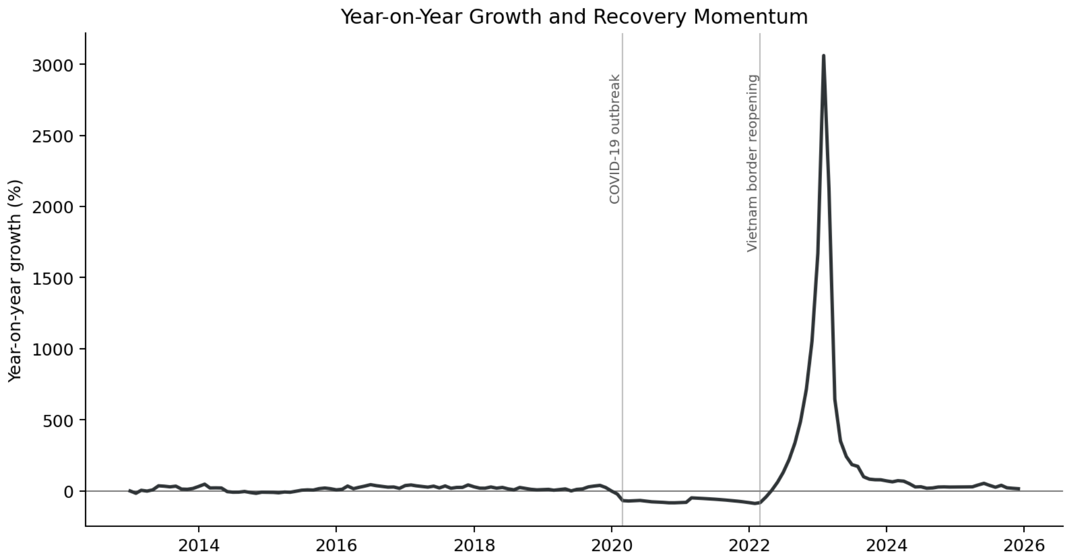

Figure 21. Month-on-month growth rates of international arrivals, 2012–2025. The COVID collapse and reopening rebound generate extreme positive and negative growth rates substantially outside the pre-pandemic volatility regime.

The growth-rate distribution confirms the presence of substantial structural instability during the pandemic and reopening periods and provides additional motivation for volatility-aware forecasting frameworks.

B Statistical Diagnostics

This appendix reports supplementary statistical diagnostics supporting the empirical findings discussed in the main text.

B.1 Stationarity and Structural Break Tests

Table 9 reports the complete stationarity test results for the log-transformed arrivals series.

Table 9. Stationarity Tests for Log International Arrivals

Series	ADF Statistic	ADF p-value	KPSS p-value
Log level	—	0.353	≥ 0.10
First difference	—	6.4×10^{-14}	≥ 0.10

Table 10 reports the endogenous structural-break test results.

Table 10. Zivot–Andrews Structural Break Test

Statistic	Value
Test statistic	-3.14
Detected break date	February 2020
Model specification	Break in level and trend
Interpretation	COVID structural break detected

B.2 Seasonal Stability Diagnostics

Table 11 reports STL-based seasonal and trend strength metrics.

Table 11. Seasonal and Trend Strength Metrics

Sample	Seasonal Strength (F_S)	Trend Strength (F_T)
Pre-COVID (2012–2019)	0.74	0.97
Full sample (2012–2025)	0.06	—

Table 12 reports Kruskal–Wallis seasonal stability tests comparing the pre-COVID and post-reopening periods.

Table 12. Seasonal Stability Tests by Calendar Month

Month	Statistic	p-value
January	—	> 0.10
February	—	> 0.10
March	—	> 0.10
April	—	> 0.10
May	—	> 0.10
June	—	> 0.10
July	—	> 0.10
August	—	> 0.10
September	—	> 0.10
October	—	> 0.10
November	—	> 0.10
December	—	0.089

B.3 Residual Diagnostic Tests

Table 13 summarises the residual diagnostic tests for all estimated forecasting models.

Table 13. Residual Diagnostic Summary

Model	Test	Statistic	p-value
SARIMA	Ljung–Box (12)	0.245	1.000
SARIMA	ARCH-LM (12)	6.782	0.872
SARIMA	Jarque–Bera	83,931	0.000
Holt-Winters	Ljung–Box (12)	9.762	0.637
Holt-Winters	ARCH-LM (12)	6.399	0.895
Holt-Winters	Jarque–Bera	304.3	0.000
SARIMA-GARCH	Ljung–Box (12)	29.861	0.003
SARIMA-GARCH	ARCH-LM (12)	1.524	1.000

Figure 22 provides supplementary evidence on the evolution of tourism-demand volatility.

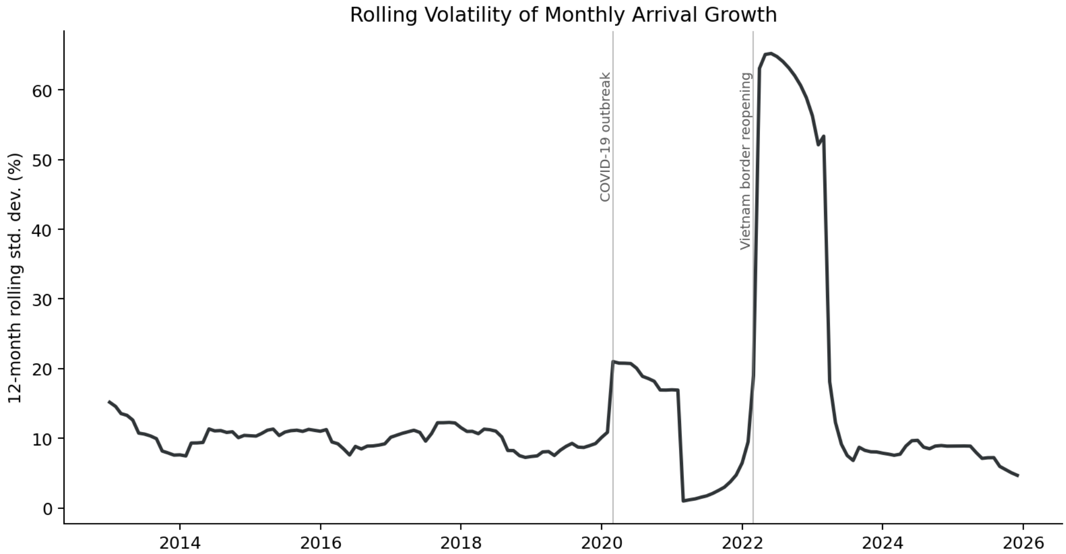

Figure 22. Rolling 12-month standard deviation of monthly tourism growth rates. The COVID collapse generates an extreme volatility regime that gradually normalises during the later recovery period.

C Model Estimation and Configuration

C.1 Forecasting Experiment Design

Table 14 summarises the forecasting experiment configuration used throughout the study.

Table 14. Forecasting Experiment Configuration

Item	Specification
Frequency	Monthly
Sample period	January 2012 – December 2025
Training window	January 2012 – December 2022
Test window	January 2023 – December 2025
Forecast horizon	36 months
Forecast design	Fixed-origin out-of-sample
Transformation	Natural logarithm
Seasonal periodicity	12 months
Primary target	International arrivals
Evaluation metrics	MAE, RMSE, MAPE, sMAPE

C.2 Model Configuration Summary

Table 15 reports the implementation details for all forecasting models.

Table 15. Model Configuration Summary

Model	Library	Key Configuration
Holt-Winters	`statsmodels`	Additive trend and seasonality; $s = 12$
SARIMA	`statsmodels SARIMAX`	AIC grid search; $d = 1$, $D = 1$, $s = 12$
SARIMA-GARCH	`statsmodels` + `arch`	GARCH(1,1) fitted on SARIMA residuals
XGBoost	`xgboost 2.x`	250 trees; max depth = 3; learning rate = 0.04

C.3 XGBoost Feature Construction

Table 16 summarises the feature groups used in the XGBoost forecasting framework.

Table 16. XGBoost Feature Set Summary

Feature Group	Features	Notes
Autoregressive lags	$y_{t-1}, y_{t-2}, y_{t-3}, y_{t-6}, y_{t-12}$	No future leakage
Rolling means	3-, 6-, and 12-month windows	Shifted by one period
Rolling standard deviations	3-, 6-, and 12-month windows	Shifted by one period
Calendar features	Month, quarter, year, time index	Deterministic
COVID period	March 2020 – February 2022	Binary indicator
Border reopening	From March 2022	Binary indicator
Recovery regime	From January 2023	Binary indicator
Russia–Ukraine war	From February 2022	Contextual event feature
China–US tensions	From July 2018	Contextual event feature
Iran–Israel conflict	From April 2024	Contextual event feature

C.4 Data Validation and Cleaning Summary

Table 17 summarises the preprocessing procedures applied to the VNAT tourism dataset.

Table 17. Data Validation and Cleaning Summary

Issue	Count	Treatment
Missing monthly observations	13	Deterministic interpolation
Repeated stale vectors	24	Corrected using adjacent temporal structure
Date formatting inconsistencies	Multiple	Standardised to monthly index
Duplicate entries	Checked	Removed where necessary

The preprocessing pipeline was designed to preserve temporal consistency while minimising artificial distortion of the underlying tourism-demand dynamics.